

Which methods for detecting AI-generated text —statistical detection, watermark

1906.04043

Abstract

Introduction

Intro

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The rapid proliferation of large language models has made machine-generated text nearly indistinguishable from human writing, creating an urgent need for reliable detection methods —yet humans alone detect fake text at barely above chance (54.2%), as documented in the earliest systematic study of this problem (arXiv:1906.04043). This survey examines the linguistic signatures that enable detection, the tools built to exploit them, and the critical limitations that persist: while GLTR demonstrated that distributional features (e.g., real text uses tail-of-distribution words $2.41\times$ more often than generated text) can boost human accuracy to 72% and classifier performance substantially, large-scale evaluations reveal that no single detector generalizes reliably across models, domains, or languages (arXiv:2306.15666), and many detectors encode systematic demographic bias —most notoriously against non-native English writers (arXiv:2304.02819). **Thesis:** This review synthesizes the empirical foundations of AI-generated text detection —from the distributional regularities first quantified by Gehrmann et al. (2019) through the failure modes and fairness concerns exposed by subsequent benchmarks —to map the current frontier and identify open challenges for robust, equitable detection.

The survey proceeds as follows: **§2** reviews the linguistic and statistical foundations of detection, centering on the distributional asymmetries established by GLTR; **§3** surveys the landscape of automated detection tools and their comparative performance; **§4** critically examines the bias, fairness, and generalization failures documented across benchmarks; and **§5** synthesizes open problems and directions for future work.

本引言综述聚焦于一个核心问题：1906.04043 (GLTR) 究竟确立了什么？该研究建立了两个关键实证发现——(1) 人类在无辅助条件下辨别机器生成文本的准确率仅为 54.2%，接近随机水平；(2) 通过分布性统计特征可视化，可将人类检测准确率提升至 72%，其核心依据是真实文本使用尾部分布词汇的频率是 GPT-2 生成文本的 2.41 倍。后续研究进一步揭示了该领域的深层挑战：大规模基准测试表明尚无单一检测器能跨模型、跨领域可靠泛化 (arXiv:2306.15666)，且检测器普遍存在对非英语母语者的系统性偏见 (arXiv:2304.02819)。本综述从 GLTR 奠定的分布规律出发，梳理检测工具的演进、公平性困境及开放问题。

The file's JSON claims got truncated at 2000 chars. Let me pull the full content via shell.## Overview
GLTR (arXiv:1906.04043) 确立了两项基础事实。其一，分布学信号：真实文本使用模型前 100 个预测之外 (长尾) 词汇的频率，是 GPT-2 生成文本的 2.41 倍 (BERT 下为 1.67 倍) (arXiv:1906.04043)。其二，其分布特征在区

分真实与生成文本上优于词袋特征，无论分类器是否接触真实生成模型 (arXiv:1906.04043)。该方法的核心主张是“人在回路”式增强：无叠加界面时，参与者检测假文本的准确率仅 54.2%，略高于随机水平 (arXiv:1906.04043)；而加入 GLTR 标注叠加后，未经任何训练即可将准确率从 54% 提升至 72% (arXiv:1906.04043)。

GLTR establishes two foundational results for statistical detection: real text uses tail-of-distribution words (outside the top-100 model predictions) $2.41\times$ more often than GPT-2-generated text ($1.67\times$ under BERT), and these distributional features separate real from generated text better than bag-of-words features (arXiv:1906.04043). Its central promise is human-in-the-loop augmentation: unaided participants detected fake text at just 54.2% accuracy, barely above chance, while the GLTR annotation overlay raised accuracy to 72% with no prior training (arXiv:1906.04043).

数年后，建基于此类思路的工具并未兑现这一乐观预期。Weber-Wulff 等判定现有检测工具“既不准确也不可靠”，且偏向将输出判为人类写作 (arXiv:2306.15666)；Pegoraro 等发现最有效的在线工具对 ChatGPT 文本的成功率仍低于 50% (arXiv:2306.15666)，van Oijen 测得工具整体准确率仅 27.9% (arXiv:2306.15666)。相反方向上，Liang 等呈现在平均 61.22% 的假阳性率下将 TOEFL 作文误判为“AI 生成”，且 97.80% 的作文被至少一个检测器标红 (arXiv:2304.02819)；增强语言多样性可将假阳性率降至 11.77%，一个二次自编辑提示词即可将 ChatGPT 文本检出率从 100% 降至 13% (arXiv:2304.02819)。

This optimism did not transfer to the automated tools built on such ideas. Weber-Wulff et al. concluded that available tools are neither accurate nor reliable and are biased toward classifying output as human-written (arXiv:2306.15666)；Pegoraro et al. found the most effective online tool fell below 50% on ChatGPT text (arXiv:2306.15666)，and van Oijen measured an overall tool accuracy of 27.9% (arXiv:2306.15666). In the opposite direction, Liang et al. found GPT detectors flagged TOEFL essays as AI-generated at an average 61.22% false-positive rate, with 97.80% flagged by at least one detector；enhancing linguistic diversity cut false positives to 11.77%，and one self-edit prompt dropped ChatGPT detection from 100% to 13% (arXiv:2304.02819).

两条证据线对偏置方向的判断相互矛盾：Weber-Wulff 认为工具偏向“人类”，Liang 则指出对非母语写作者偏向“AI”——但两者都同意检测的脆弱性。留下的开放缺口是：GLTR 在 GPT-2 时代（早于 ChatGPT 两月破亿订阅者的井喷，arXiv:2306.15666）验证的 54%→72% 人工检测增益，是否仍适用于现代 LLM 输出，且该增益是否重现当前自动化检测器的偏置失败。

The two strands explicitly disagree on bias direction—tools skew toward “human” in Weber-Wulff, toward “AI” for non-native writers in Liang—yet agree that detection is fragile. The open gap: whether GLTR’s distribution-augmented human gain (54%→72%), demonstrated on GPT-2-era text before ChatGPT’s 100-million-subscriber surge (arXiv:2306.15666), transfers to modern LLM output, and whether it reproduces the bias failures of today’s automated detectors.

The file is truncated. Let me get the full content.## Conclusion

This survey asks what arXiv:1906.04043 establishes—and the answer is a foundationally useful but historically bounded result: machine-generated text, at least as produced by GPT-2-era models, carries systematic statistical fingerprints that both machines and humans can exploit.

Key takeaways. First, detection is not hopeless for humans *when* they are given the right cues. Feeding participants GLTR’s distributional overlay lifted their fake-text detection from 54.2%—barely above chance—to 72% accuracy without any training (arXiv:1906.04043). Second, the underlying signal is real and quantifiable: real text samples the tail of the model’s predictive distribution (words outside the top-100 predictions) $2.41\times$ more often than generated text under GPT-2 ($1.67\times$ under BERT), and these distributional features separate real from generated text better than bag-of-words features, with or without access to the true generating model (arXiv:1906.04043). Third, that discriminative signal decays with model generations and in the wild. Weber-Wulff et al. report that deployed tools reached ~27.9% overall accuracy, with the best tool capping at 50%, and being biased toward calling output human-written (arXiv:2306.15666)；Liang et al. show GPT detectors flag 97.8% of TOEFL essays as AI-generated

by at least one detector (61.22% average false-positive rate), while a second-round self-edit prompt cut ChatGPT essay detection from 100% to 13% (arXiv:2304.02819). Taken together, the three sources tell one coherent story: statistical detectability is a property of a *specific model family at a specific time*, not a stable property of “AI text” as a class.

Remaining gaps. (1) GLTR’s 54.2%→72% result and its $2.41\times$ tail-word ratio are calibrated to GPT-2/BERT; the survey lacks an equivalent human-subject and feature-validity replication against post-2023 instruction-tuned and reasoning models, where distributional gaps shrink. (2) The statistics-vs-classifier thread (GLTR features beating bag-of-words) is not connected to the tool-evaluation thread (Weber-Wulff/Liang) —a systematic comparison of statistical feature methods against deployed detectors on the same benchmark corpus is missing. (3) The bias results are domain-limited (English academic/TOEFL text); performance on non-English, code, and multilingual text remains *unverified* in the cited sources. (4) No source addresses the adversarial equilibrium: as detectors improve, generators and simple prompts (self-edit) adapt, so static accuracy figures age quickly.

Bottom line. 1906.04043 establishes that distributional deviations from a known language-model prior are measurable and usable —the *methodological basis* for the entire detection craft. But the later literature (arXiv:2306.15666; arXiv:2304.02819) establishes the *ceiling*: without the true generating model —and with bias risks to innocent writers —any claim of reliable, fair detection must be treated as provisional.

结论 (中文摘要)

本调研试图回答: arXiv:1906.04043 到底确立了什么。答案是——一个奠基性但**历史有界**的结论: 机器生成文本 (至少是 GPT-2 时代的模型) 带有系统性统计指纹, 机器与人类都可利用。

核心要点。其一, 只要提供正确线索, 人类检测并非无望: GLTR 的分布叠加界面让被试对伪造文本的识别率从 54.2% (仅略高于随机) 提升至 72%, 且无需任何事前训练 (arXiv:1906.04043)。其二, 底层信号真实且可量化: 真实文本在模型预测分布尾部的采样频率 (top-100 之外) 是生成文本的 2.41 倍 (GPT-2 下; BERT 下为 1.67 倍), 且无论是否接入真实生成模型, 这些分布特征对真伪文本的区分度都优于词袋特征 (arXiv:1906.04043)。其三, 该判别信号随模型代际更替而在实际场景中衰减: Weber-Wulff 等的工具测试显示总体准确率仅约 27.9%、最佳工具封顶 50%, 且偏向于把输出判为人类写作 (arXiv:2306.15666); Liang 等发现 GPT 检测器将 97.8% 的 TOEFL 作文至少在一种检测器下标记为 AI 生成 (平均误报率 61.22%), 而一轮“二次自编辑”提示即可把 ChatGPT 作文检出率从 100% 打到 13% (arXiv:2304.02819)。三份文献共同讲出一个连贯的故事: **统计可检测性是“特定模型族在特定时间”的属性, 而非“AI 文本”这一类别的稳定属性。**

尚存空白。(1) GLTR 的 54.2%→72% 结果与 2.41 倍尾词比率均以 GPT-2/BERT 标定, 缺少针对 2023 年后指令微调与推理模型的人类被试与特征有效性复制实验——此时分布间隙已收窄。(2) “统计特征 vs 分类器”线索 (GLTR 优于词袋) 与“工具评测”线索 (Weber-Wulff/Liang) 未有衔接: 缺少在同一基准语料上对统计特征方法与已部署检测器的系统对比。(3) 偏差结论局限于英语学术/TOEFL 文本; 非英语、代码与多语文本的表现在前述文献中仍标记为 *unverified*。(4) 没有文献处理对抗均衡: 检测器进步的同时, 生成器与简单提示 (自编辑) 也在演化, 静态准确率数字快速过时。

一句话总结。1906.04043 确立了“已知语言模型先验下的分布偏离”可测量、可利用——这是整个检测方法论的地基; 而后续文献 (arXiv:2306.15666; arXiv:2304.02819) 确立了其**上限**——一旦缺乏真实生成模型、并对无辜写作者存在误报风险, “可靠且公平的检测”这一声称都只能被当作暂时性结论。

Evidence Table

#	Claim (abridged)	Source	Confidence
1	GLTR improves human detection of fake text from 54% to 72% accuracy without prior training	arXiv:1906.04043	high
2	Real text uses tail-of-distribution words $2.41\times$ more frequently than generated text under GPT-2	arXiv:1906.04043	high
3	Without GLTR overlay, participants detected fake text at only 54.2% accuracy, near random chance	arXiv:1906.04043	high
4	GLTR distributional features outperform bag-of-words features for separating real from generated text	arXiv:1906.04043	high
5	Within two months of its launch, ChatGPT had over 100 million subscribers and was labelled the fastest growing consumer ...	arXiv:2306.15666	high
6	Pegoraro et al. claimed that the most effective online detection tool could only achieve a success rate of less than 50%...	arXiv:2306.15666	high
7	van Oijen’ s tests showed that the overall accuracy of tools in detecting AI-generated text reached only 27.9%, and the b...	arXiv:2306.15666	high
8	The study concludes that available detection tools are neither accurate nor reliable and have a main bias towards classi...	arXiv:2306.15666	high
9	GPT detectors misclassified over half of TOEFL essays as AI-generated with an average false positive rate of 61.22%	arXiv:2304.02819	high
10	97.80% of TOEFL essays were flagged as AI-generated by at least one detector	arXiv:2304.02819	high
11	Enhancing word choices to mimic native speakers reduced the false positive rate from 61.22% to 11.77%	arXiv:2304.02819	high
12	A second-round self-edit prompt significantly reduced ChatGPT essay detection rates from 100% to 13%	arXiv:2304.02819	high

References

- [\[1\]](#) arXiv:1906.04043
- [\[2\]](#) arXiv:2306.15666
- [\[3\]](#) arXiv:2304.02819